

Questions & Analytical Rationale

Section A • Basic & General Customer Profiling (Questions 1–5)

Q1. Loyal vs. Discount-Only Customers

- Establishes a baseline understanding of how broad customer segments and promo dependency groups correlate with overall loyalty and spend.

Q2. Behavioral Patterns Predicting High Value

- Identifies the basic behavioral metrics (past purchases, frequency, subscription scores) that differentiate high-tier customers from low-tier ones.

Q3. Geographically Underlevered Markets

- Provides a simple regional breakdown of customer count and spend to find states with high concentrations of premium customers.

Q4. Restructuring the Promotional Strategy

- Evaluates the baseline impact of discounts by grouping customers by their promotional dependency tier and value tier.

Q5. Ideal Customer Profile (ICP) Extraction

- Filters for top-tier, non-promo-dependent customers to outline the basic demographic and preference traits of the ideal buyer.

Section B • Advanced Business Problem Solving (Questions 6–15)

Q6. Promo Margin Leakage & Profit-at-Risk Analysis

- Identifies locations where extreme discount dependency is draining revenue without generating actual customer loyalty, highlighting areas to reduce promotional spend.

Q7. Cross-Sell & Category Expansion Matrix

- Assesses subscription penetration across top product categories to design targeted bundle offers and cross-sell campaigns.

Q8. "At-Risk" High-Value Churn Prediction Profiler

- Isolates high-margin customers who are exhibiting churn signals (poor review ratings or low satisfaction flags) so the retention team can intervene before revenue is lost.

Q9. Cumulative Revenue Concentration (Decile Pareto Analysis)

- Uses advanced window functions (NTILE) to mathematically divide the customer base into deciles, calculating the exact threshold for VIP loyalty tiers.

Q10. Multi-Channel Preference Optimization

- Cross-tabulates payment and shipping methods to discover which operational combinations yield the highest retention and lower transaction friction.

Q11. Segment Benchmark Variance Analysis (Multi-Table Join)

- Joins granular data with segment_analysis to find "overachievers" — individuals in lower-tier segments who are outperforming their segment averages by 50% and should be upgraded.

Q12. Geographic Efficiency vs. State Benchmark Comparison (Multi-Table Join)

- Joins data with geography_analysis to identify exceptionally loyal customers who live in heavily promo-dependent states, proving that brand loyalty can survive in discount-heavy markets.

Q13. Category Expectation vs. Reality Check (Multi-Table Join)

- Compares an individual's review rating against the global average in category_analysis to flag customers experiencing significant satisfaction deficits in otherwise high-performing categories.

Q14. The Subscription Revenue Premium Analysis

- Quantifies the exact revenue and frequency gap between subscribed and non-subscribed customers within the same category to justify budget for a "Subscribe & Save" marketing push.

Q15. High-Frequency Shipping Drain Identification

- Isolates "margin killers" — customers who order extremely frequently using premium shipping (Express) while relying entirely on massive discounts, helping logistics restructure free-shipping thresholds.